# Oluwatosin Ayeni

Staff Software Engineer, Inference — Anthropic

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## PROFILE

Staff-calibre AI Inference Engineer with expertise in distributed LLM serving, GPU optimisation, batching pipelines, and Kubernetes GPU infrastructure. Benchmarked H100 NVL — 5,965 tok/s peak, 1,443× TTFT speedup, 100% SLO compliance at concurrency 128. Built production inference gateways (3.87× faster, 14× throughput) and Kubernetes LLM serving stacks. Deep alignment with Anthropic's mission — I use Claude in production and build the same infrastructure layer that serves it.

## CORE COMPETENCIES

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|---|---|
| **Distributed Inference** | Tensor parallelism TP=2/4, pipeline parallelism, KV cache-aware routing, multi-node GPU orchestration, batching strategies (continuous, dynamic, chunked prefill) |
| **Inference Engines** | vLLM, SGLang, TensorRT-LLM, ONNX Runtime — benchmarking, profiling, production deployment |
| **GPU & CUDA** | NVIDIA Nsight Systems/Compute, memory bandwidth profiling, kernel fusion, warp occupancy, BFloat16/FP8 |
| **ML Infrastructure** | Kubernetes (HPA, GPU node pools), Docker, Prometheus, Grafana, FastAPI, Supabase, GitHub Actions, Python |
| **LLM Techniques** | Speculative decoding, prefix/RadixAttention caching, AWQ/GPTQ quantisation, flash attention, PEFT/LoRA, multi-provider routing |

## PROFESSIONAL EXPERIENCE

Jan 2025–present **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

## KEY PROJECTS

**llm-inference-benchmark (H100)** 2× H100 NVL 94 GB, CUDA 13.4. vLLM v0.30 vs SGLang v0.5.20, Qwen3-8B BF16, TP=2. RadixAttention: 1,443× TTFT speedup. Peak: 5,965 tok/s. SLO 100% at concurrency 128. Live results

**inference-optimizer** Production Inference Gateway: multi-provider (Claude, GPT, Gemini), 3.87× faster, 14× throughput, 61.4% cost reduction. GitHub

**vllm-k8s-stack** Kubernetes LLM serving: vLLM, KV-cache-aware routing, HPA autoscaling, Prometheus/Grafana. GitHub

**inference-harness** Unified benchmark harness: Anthropic, OpenAI, Gemini + vLLM, SGLang, TensorRT-LLM — GPU profiling, automated reporting. GitHub

**open-model-engine** GPU-first benchmarking: CUDA profiling, quantisation, speculative decoding. GitHub

## EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

## OPEN SOURCE

**GitHub** 43 public repos — github.com/AyeniOluwatosinOlawale